

Notice firstly that $\alpha_j + \gamma_j$ separates the surface S , meaning that $\alpha_j + \gamma_j = 0 \in H_1(S)$, or in other words,

$$-\alpha_j = \gamma_j = \tau_*(\alpha_j), \quad (4.3)$$

where τ_* denotes the automorphism of $H_1(S)$ induced by τ , i.e. $\tau_* = \Psi(\tau)$. Notice secondly that β_j was sent to its inverse, or in other words,

$$-\beta_j = \tau_*(\beta_j). \quad (4.4)$$

We've shown that equations (4.3) and (4.4) hold for all odd j inside $\{1, \dots, n\}$. Now, by braid and disjointness relations, we have that

$$cbabcb = cbacbc = cbcabc = bcbabc,$$

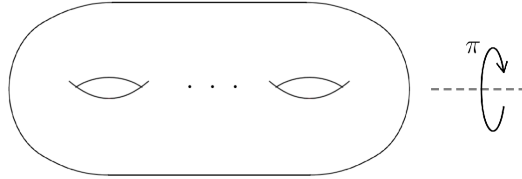
but then, since $cbabcb = \tau$ has order 2, we know $cbabcb$ is equal to its own inverse, and thus

$$cbabcb = (bcbabc)^{-1} = c^{-1}b^{-1}a^{-1}b^{-1}c^{-1}b^{-1}.$$

This shows that every right Dehn twist in $(cab)^2$ can be replaced with a left Dehn twist and vice versa, and nothing will change. Therefore, if $j \in \{1, \dots, n\}$ is even, $(cab)^2$ will affect the curve α_j and β_j in the exact same manner as what is depicted in Figure 6 and Figure 7. We conclude that equations (4.3) and (4.4) hold for all $j \in \{1, \dots, n\}$ and hence, $\Psi(\tau) = \tau_* = -\text{Id}_{2n} \in \text{Sp}_{2n}(\mathbb{Z})$, as required. $\square$

The proof above can be utilized to prove a slightly more precise corollary which will come in handy in the next section. For the following result, we let λ denote the homeomorphism of S given by the following rotation by 180° :

FIGURE 8



Corollary 4.2 (Corollary of the previous proof). *Let S be a closed surface of genus n and let $[\lambda] \in \text{Mod}(S)$ denote the mapping class of λ . We have that $[\lambda] = (cab)^2 \in \Theta_n^1$.*

Proof. Consider the following simple closed curves in S , which satisfy the assumptions required for the Alexander method [1, Section 2.3]:

FIGURE 9

